

# FRONT-END WEB DEVELOPMENT

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# LEARNING OUTCOMES

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01

Demonstrate proficiency of the basic building blocks of a front end web application development framework / library.

02

Demonstrate proficiency of the programming constraints that apply when developing front end web applications.

03

Design, develop and deploy a stand alone front end web application using an Integrated Development Environment (IDE).

04

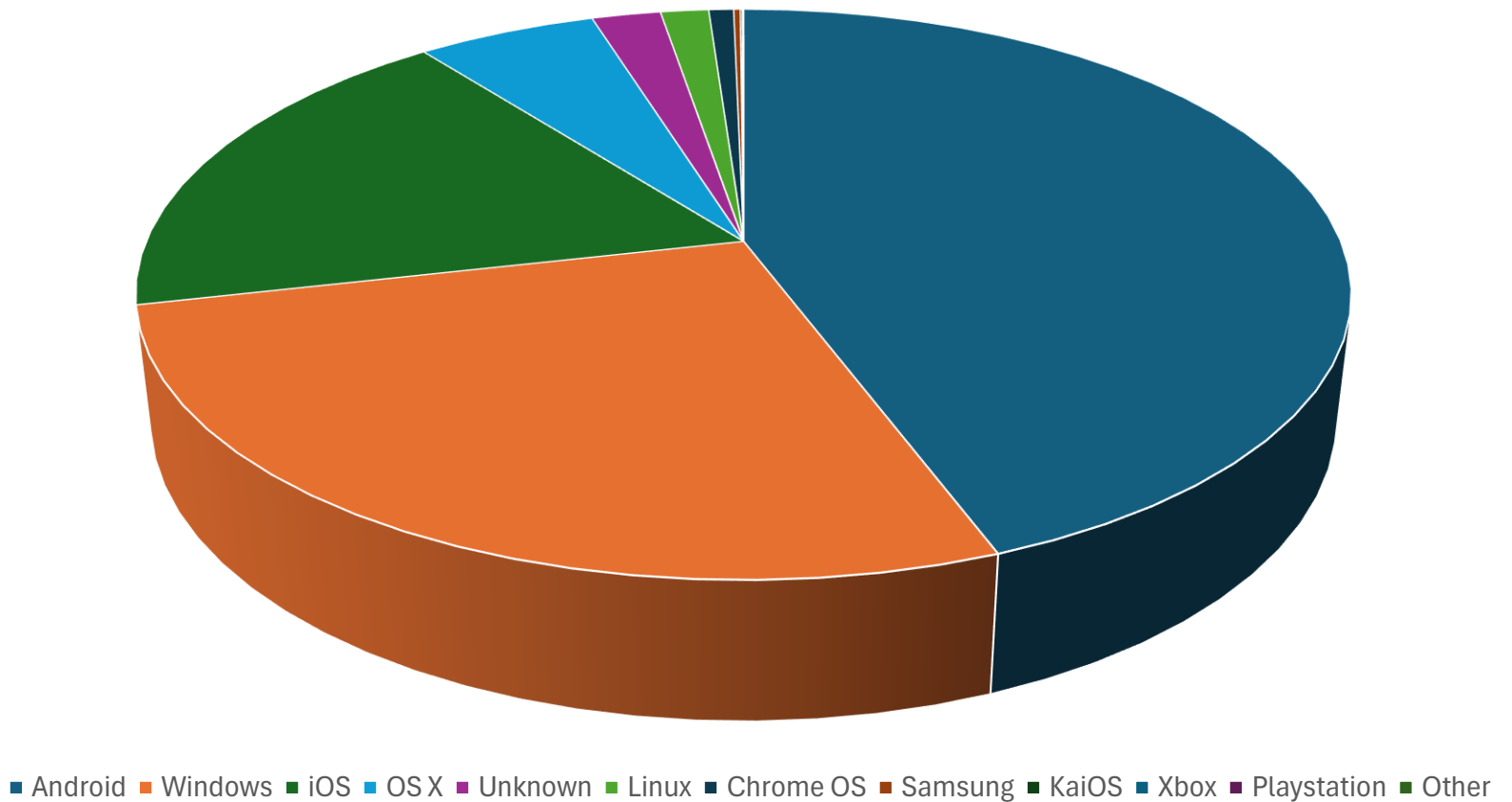
Recognise and employ real-world best practices in developing front end web applications.

# PC / TABLETS / SMART PHONES



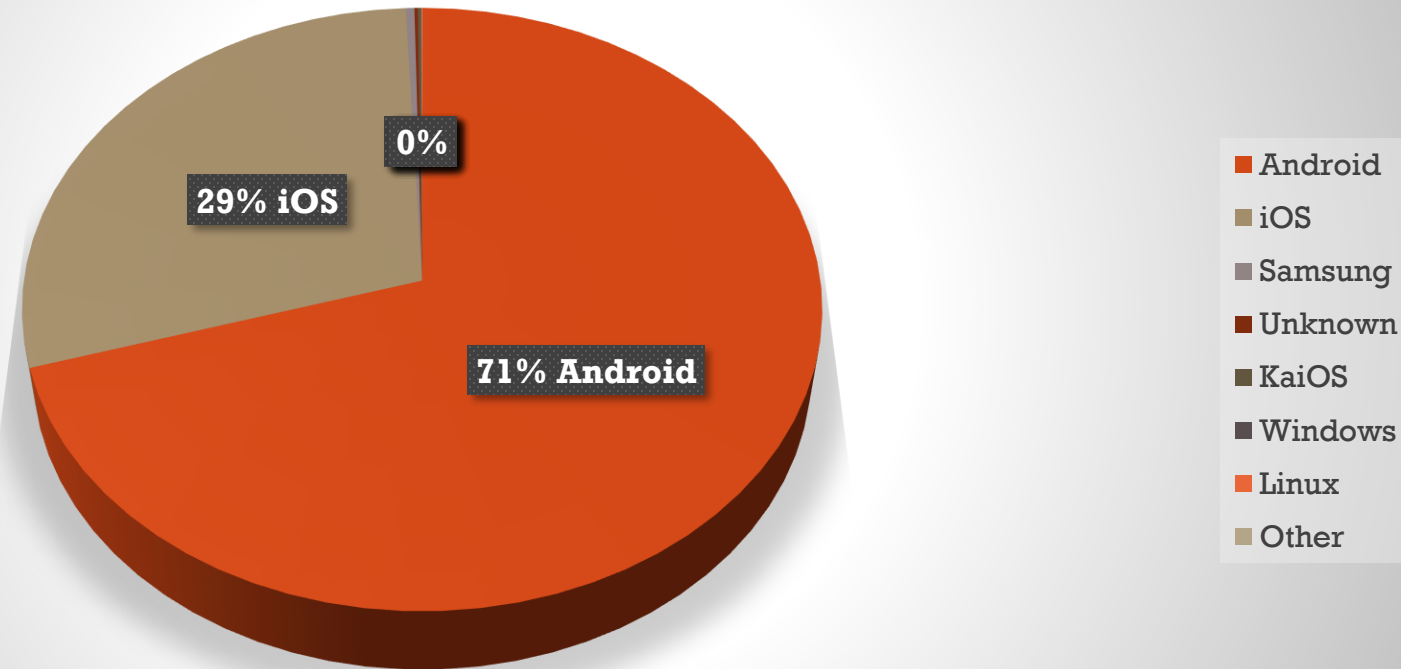
# OS MARKET SHARE 2004

| OS          | Market Share Perc. (Dec 2023 - Dec 2024) |
|-------------|------------------------------------------|
| Android     | 44.28                                    |
| Windows     | 27.16                                    |
| iOS         | 18.03                                    |
| OS X        | 5.69                                     |
| Unknown     | 2.19                                     |
| Linux       | 1.53                                     |
| Chrome OS   | 0.79                                     |
| Samsung     | 0.21                                     |
| KaiOS       | 0.06                                     |
| Xbox        | 0.02                                     |
| Playstation | 0.01                                     |
| Other       | 0.02                                     |



# SMART PHONE SALES BY OS

Market Share Perc. (Dec 2022 - Dec 2023)



# APPLICATION DEVELOPMENT OPTIONS



# NATIVE



- Native apps are apps that have been developed for use on a particular platform.
  - For example, Swift for iOS, Java for Android.
- Advantages
  - Access to the latest APIs
  - Access to hardware
  - Most Responsive
  - Best looking
- Disadvantages
  - Won't work on other platforms
  - Steep learning curve
  - Higher development cost, if targeting many platforms



# NATIVE APPLICATIONS



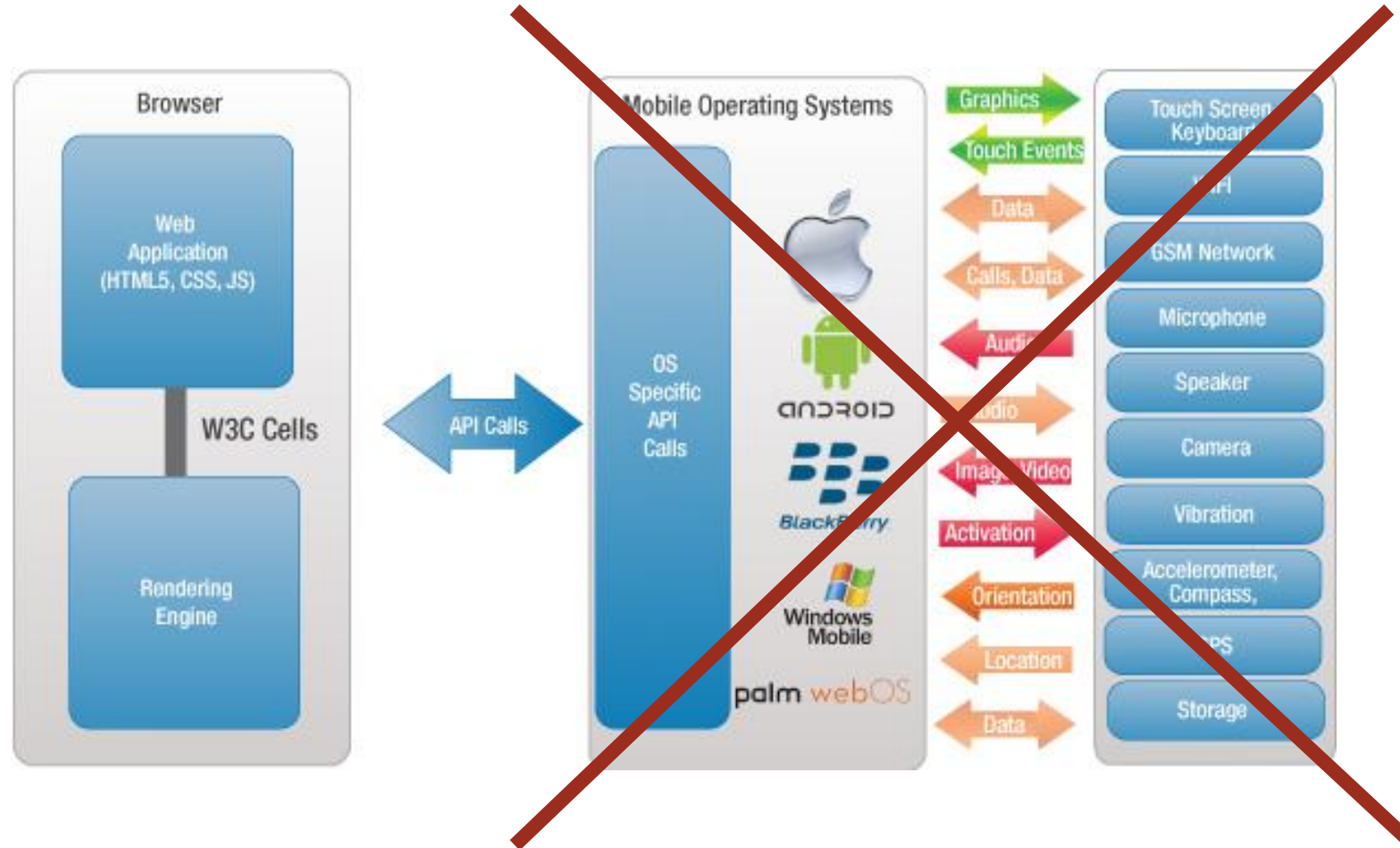


# WEB



- Runs in a web browser, not on the mobile platform's OS.
- **Advantages**
  - Use standard web technologies e.g. HTML5, CSS and JavaScript
  - Write-Once-Run-Anywhere
  - Not downloaded from App Store or Google Play
- **Disadvantages**
  - Since they don't run from the OS they have limited access to system resources
  - Therefore not suitable for developing high end apps that require such access

# WEB APPLICATIONS



# HYBRID



- Runs in a “container” that provides a consistent set of JavaScript APIs to access device capabilities through plug-ins, which are built with native code.

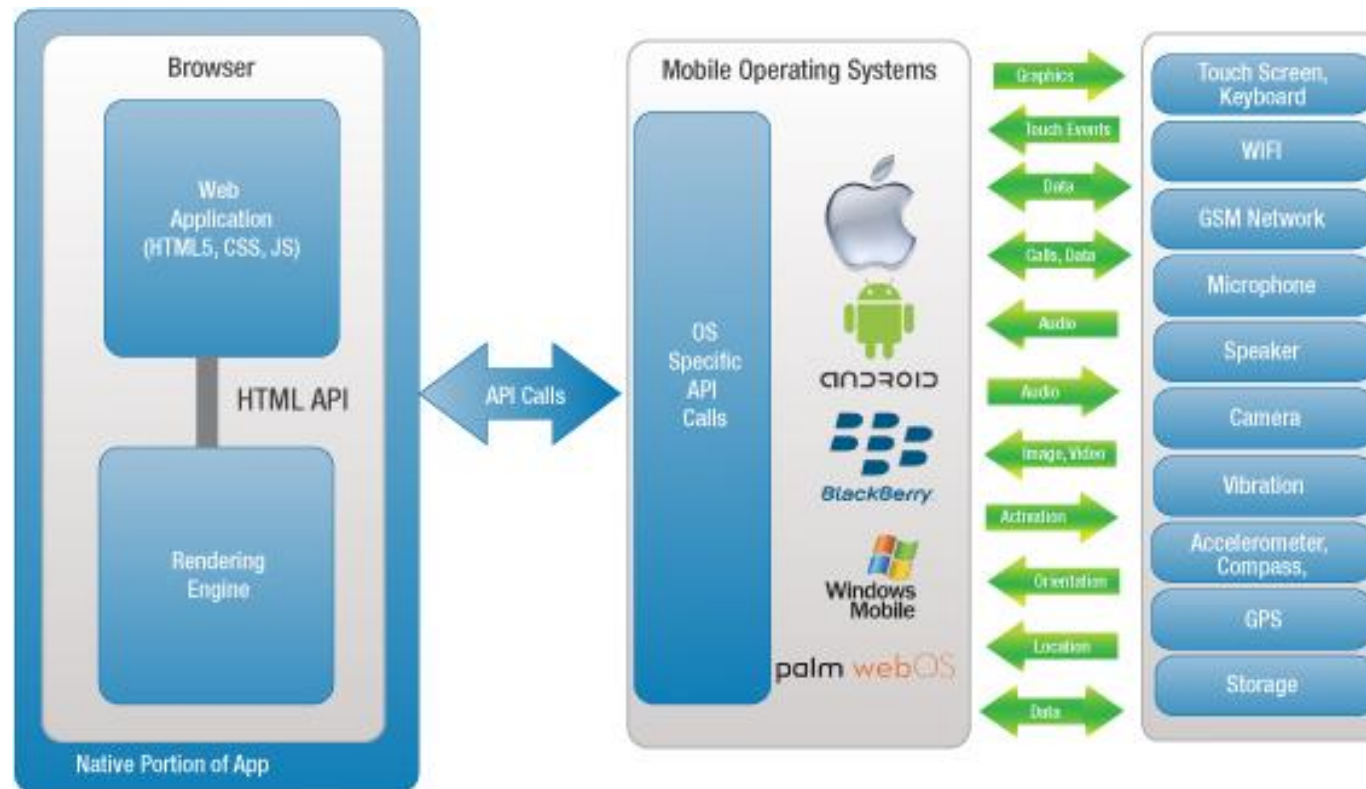
- Advantages

- Use standard web technologies e.g. HTML5, CSS and JavaScript
- More functional than web apps, as can access device’s capabilities
- Deployable across many platforms

- Disadvantages

- Still not as responsive as native apps
- The newest API features will not be available immediately

# HYBRID APPLICATIONS



# WHICH OPTION WILL WE USE?

- Hybrid
- Why?
  - Productivity is much faster than native
  - Standard web languages HTML5, CSS, JavaScript (**Angular**)
- Will be able to develop your own apps which you can put on your own device.



# Progressive Web Apps



# HYBRID PLATFORMS



Xamarin



APACHE

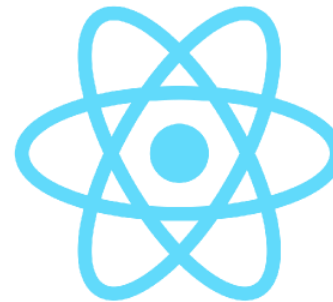
CORDOVA™



PhoneGap



Flutter



React

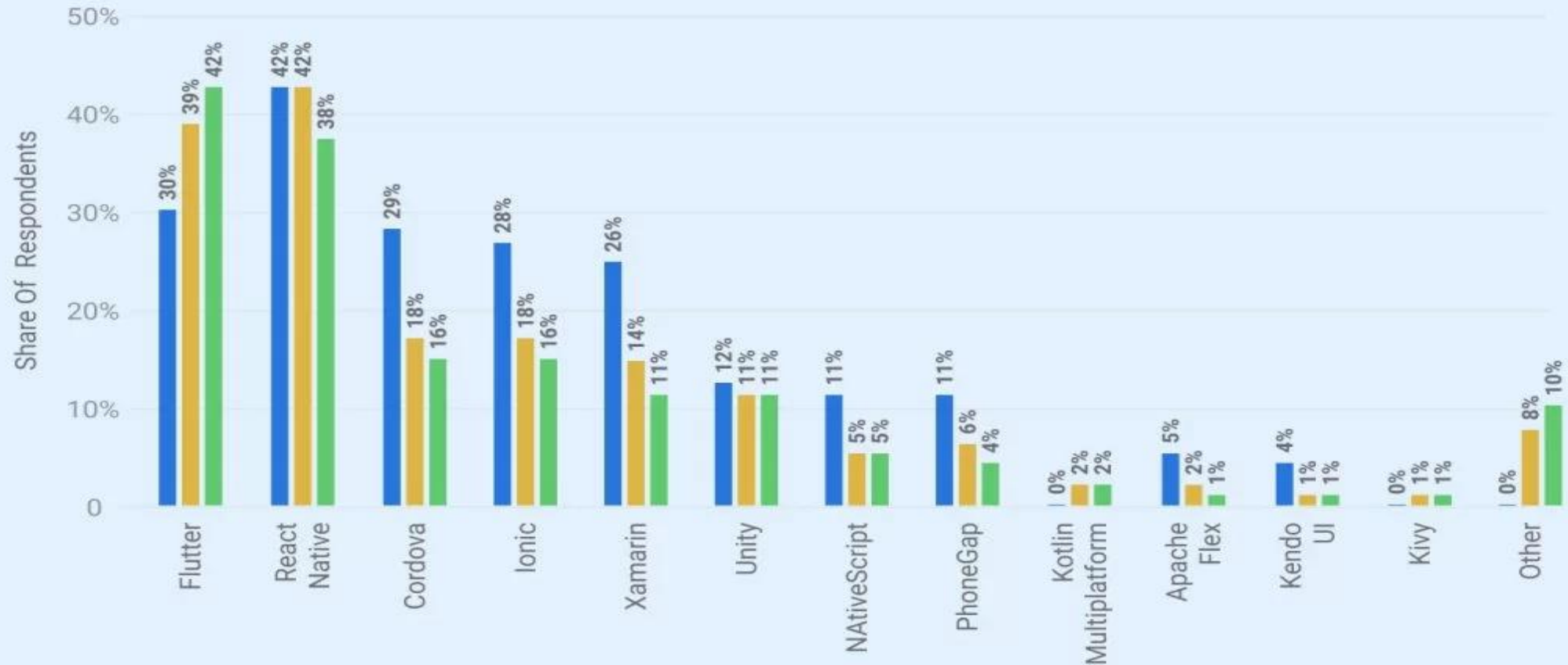


Onsen UI



# HYBRID PLATFORMS

Various Cross-Platform App Development Frameworks





# TOPICS

- Ionic 8
- Angular 21
- TypeScript
- HTML / CSS

# WHY IONIC?

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One code  
base

Angular

Native  
Plugins with  
Capacitor

Open  
Source

Big Online  
Community

# TYPESCRIPT

- Types
- Access Modifiers
- Custom Types
- Promises
- Observables
- Lambda Expression
- This

# ANGULAR

- Structure
- Components
- Data Binding
- Services
- Dependency Injection
- Observables

# IONIC

- Navigation
- Providers
- Persistent Storage
- Native Plugins

# GIT

# PREREQUISITES

Open Source Software

Node.js (<https://nodejs.org/en/>)

Git (<https://git-scm.com/>)

Visual Studio code (<https://code.visualstudio.com/>)



# STRUCTURE OF COURSE

- Lecture 2hr Wednesday 1pm – 3pm
- Labs 2hr Mondays, Wednesday, Friday



# ASSESSMENT SCHEDULE

| Assessment               | Percentage | Due Date                                       |
|--------------------------|------------|------------------------------------------------|
| <b>Typescript MCQ</b>    | 30%        | Week ~4                                        |
| <b>Angular MCQ</b>       | 30%        | Week ~9                                        |
| <b>Labs (every week)</b> | 10%        | Weeks 2,3,5,6,7,8,10,11<br>(worth ~1.25% each) |
| <b>Ionic Project</b>     | 30%        | Week 13 (VIVA)                                 |

# RECOMMENDED RESOURCES

- TypeScript: <https://www.typescriptlang.org/>
- Angular: <https://angular.dev>
- Ionic: <http://ionicframework.com/>
- TypeScript: Book – TypeScript Essentials – Available as an e-book from the library

# QUESTIONS